

S-DNA and RecA/RAD51-Mediated Strand Exchange *in Vitro*

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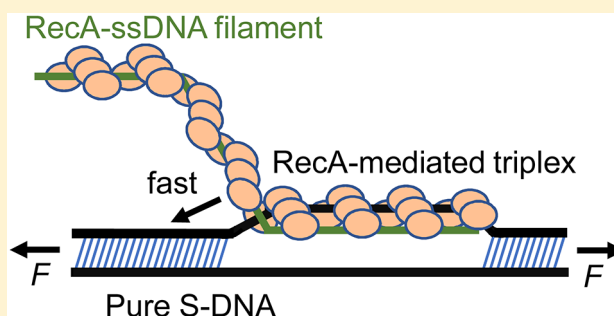
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Supporting Information

ABSTRACT: S-DNA (stretched DNA) is an elongated base-paired DNA conformation under high tension. Because the RecA/Rad51 family DNA recombinases form helical filaments on DNA and mediate the formation of the DNA triplex (D-loop), in which the DNA is stretched, and because the extension of these nucleoprotein filaments is similar to the extension of S-DNA, S-DNA has long been hypothesized as a possible state of DNA that participants in RecA/Rad51-mediated DNA strand exchange in homologous recombination. Such a hypothesis, however, is still lacking direct experimental studies. In this work, we have studied the polymerization and strand exchange on S-DNA mediated by *Escherichia coli* RecA, human Rad51, and *Saccharomyces cerevisiae* Rad51 by single-molecule magnetic tweezers. We report that RecA/Rad51 polymerizes faster on S-DNA than on B-DNA with the same buffer conditions. Furthermore, the RecA/Rad51-mediated DNA triplex forms faster from S-DNA than from B-DNA together with the homologous single-stranded DNA. These results provide evidence that S-DNA can interact with RecA and Rad51 and shed light on the possible functions of S-DNA.



Double-stranded DNA (dsDNA) generally exists in the right-handed double helical B-DNA conformation *in vivo*. Upon stretching with a tensile force of around 65 pN, torsion-unconstrained DNA can elongate to ~ 1.7 times the B-DNA contour length, which is called DNA overstretching transition.^{1,2} The nature of DNA overstretching has been debated among three possible mechanisms:^{3–19} (i) one DNA strand peeling off at nicks or open ends to generate a single-stranded DNA (ssDNA) bearing tension while the other ssDNA relaxed, namely, peeled ssDNA; (ii) two DNA strands melting from inside to generate two ssDNA strands sharing tension, namely, bubbled DNA; and (iii) a new form of base-paired DNA, namely, S-DNA (stretched DNA). Recent studies have solved the debate on DNA overstretching by revealing all of the three hypothesized overstretched DNA structures exist but under different conditions.^{20–22} Under conditions where the base pairs are stable (e.g., GC-rich content, end-closed and torsion-unconstraint DNA topology, low temperature, and high salt), the S-DNA is preferred. Although it has not been determined by structural biology techniques, S-DNA was revealed to be a ladder-like structure using an energy minimization method.²³

In addition to DNA overstretching induced by force, B-DNA can also be elongated to ~ 1.5 times its contour length by the polymerization of RecA/RAD51 family DNA recombinases.

The RecA/RAD51 family is broadly conserved across all kingdoms of life including eukaryotic Rad51 and Dmcl1, archaeal Rada, and prokaryotic RecA. The RecA is the critical protein involved in several steps of prokaryotic homologous recombination processes. First, RecA polymerizes onto an initiating ssDNA with ATP, and then the RecA-ssDNA nucleoprotein filament searches for its homologous dsDNA. Once aligned with its homologous dsDNA, the ssDNA invades into the dsDNA and pairs with its complementary DNA strand, generating a RecA-mediated DNA triplex (D-loop). At last, RecA unbinding from the DNA triplex with the hydrolysis of ATP and the broken DNA molecule is repaired using genetic information on its homologous DNA. In eukaryotes, the Rad51 protein works in a similar way as RecA.^{24–26}

Because S-DNA (~ 0.58 nm/bp) has a similar extension to the RecA/Rad51-DNA filament as well as the RecA/Rad51-mediated DNA triplex (~ 0.51 nm/bp), S-DNA has long been hypothesized to play a role in RecA/Rad51-mediated recombination.^{20,27,28} Previous single-DNA manipulation experiments revealed that RecA polymerized on overstretched DNA even faster than on B-DNA,²⁷ and thus, DNA

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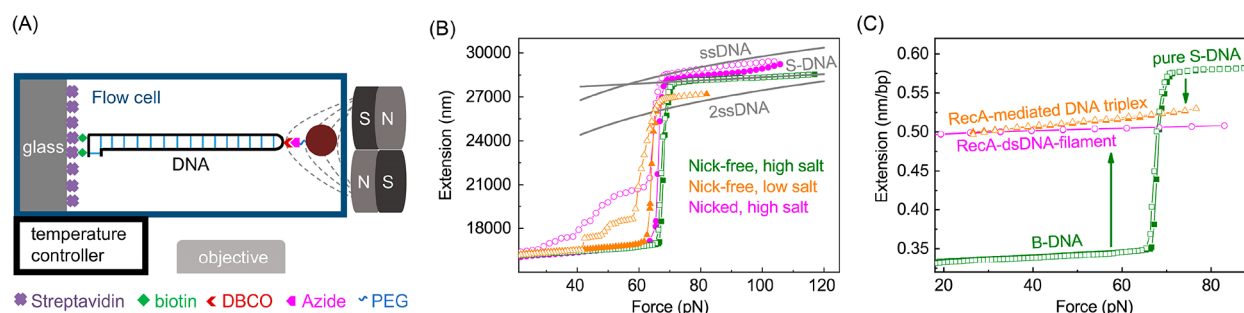


Figure 1. Experiment setup. (A) Transverse magnetic tweezers. The end-closed, torsion-unconstrained DNA molecule was stretched between a streptavidin-coated glass-slide edge and an azide-functionalized magnetic microbead. The solution and temperature inside the flow cell were controlled. (B) Determination of the nick-free DNA tethers by force–extension curves. The force–extension curves were obtained through a stretching process (filled symbols) followed by a relaxing process (empty symbols). The force–extension curve of S-DNA is fitted to a linear function of force (the gray line labeled as S-DNA). The theoretical predictions for an ssDNA strand under tension (the gray line labeled as ssDNA) and a DNA bubble containing two ssDNA strands sharing tension (the gray line labeled as 2ssDNA) were shown for comparison. (C) Comparison of the extensions among B-DNA, S-DNA, RecA-dsDNA filament, and RecA-mediated DNA triplex at 20 °C. The force–extension curves were obtained through a stretching process (filled symbols) followed by a relaxing process (empty symbols). The extension of B-DNA and S-DNA was measured at pH 5.5. The extension of the RecA-dsDNA filament was measured after sufficient incubation with 1 μM of RecA at buffer pH 5.5 containing 1 mM ATP. The extension of the RecA-mediated DNA triplex was measured after sufficient incubation with a 24 pM RecA-ssDNA filament in buffer, pH 7.5, containing 1 mM ATPγS.

overstretching was used as a way to help RecA to nucleate on dsDNA.^{29–31} However, the overstretched DNA in those experimental conditions was likely to be a mixture of peeled ssDNA formed at the DNA nicks or open ends and S-DNA formed at the GC-rich region, and thus, we still do not know whether RecA/Rad51 can polymerize on a pure S-DNA. In addition, previous magnetic tweezer experiments have revealed the real-time dynamics of the formation of the RecA-mediated DNA triplex between B-DNA and its homologous RecA-ssDNA filaments.^{32,33} However, studies on the interactions between S-DNA and its homologous RecA-ssDNA filaments are still lacking.

In this work, we design an end-closed and torsion-unconstrained DNA construct and stretch the DNA to the pure S-DNA conformation with magnetic tweezers. We study whether and how RecA/Rad51 polymerizes on the pure S-DNA at various pH values, temperatures, and DNA tensions. Furthermore, we also study whether and how the RecA/Rad51-mediated DNA triplex forms between the pure S-DNA and its homologous ssDNA during strand exchange *in vitro*.

MATERIAL AND METHODS

DNA Construct. In order to make a strong DNA construct withstand a high tension for hours, the DNA molecule was labeled with multiple biotin groups at one end and labeled with an internal DBCO group at the other covalently continuous end (Figure 1A). Such a DNA construct was end-closed and torsion-unconstrained. The 48.5k bp λ phage DNA (Roche) was used in experiments studying the dynamics of RecA/RAD51 polymerization, a 12.3k bp DNA fragment and an 8k bp GC-rich DNA fragment amplified from λ DNA were used for experiments studying the formation of RecA/Rad51-mediated DNA triplex. Please refer to Supporting Information for details of the preparation of the end-closed DNA construct (Figure S1).

Azide-Functionalized Microbeads. The azide-functionalized superparamagnetic microbeads were prepared by incubation of 100 μL of 3 μm diameter amine-functionalized silicon beads (no. 3131, Tianjin Baseline ChromTech Research Centre) with 20 mg of N3-PEG-SC, 2K (HE006023–2K,

Hunan Hua Teng Pharmaceutical Co., Ltd.) in 100 mM sodium bicarbonate solution for 2 h. The magnetic moment of this type of silicon bead is about 3-fold of that for Dynabeads M-270 (Thermo Fisher Scientific), which enables us to exert high tension on the DNA molecule.

Transverse Magnetic Tweezers. A home-built transverse magnetic tweezer setup with a Peltier temperature controller was applied in this research.^{6,34} As shown in Figure 1A, the multiple-biotin-labeled end of a DNA molecule was anchored to a streptavidin-coated glass-slide edge in a flow cell, and the DBCO-labeled end of the DNA molecule was anchored to an azide-functionalized superparamagnetic microbead. The DBCO group and azide group can form a single covalent bond without catalyst,³⁵ which enables the stretched DNA torsion-unconstrained meanwhile capable of bearing high tension.¹⁶

The extension of the stretched DNA molecule was determined by the separation between the bead and glass-slide edge. Because the microbead was very close to the glass-slide edge, sometimes the image quality was bad because of the interference of the glass-slide shadow. If the image quality was bad, although the average in DNA extension was correct, additional noise in DNA extension was generated when the location of the microbead is calculated using the gray centroid method. Thus, the noise of the time-extension courses may be larger than the thermal fluctuation of the microbead. Under the same forces of 8, 25, 45, and 75 pN, the noises in DNA extension obtained using bad images is larger than that obtained using good images (Figure S2).

A pair of NdFeB magnets was applied to attract the bead and thus exerted a constant force to the anchored DNA molecule. In order to prevent the influence of bad image quality to the calculation of the force based on the Brownian motion of the microbead,³⁶ in practice, we chose a standard bead with the best image quality to determine the magnetic force (F) as a function of the distance between the magnets and the microbead (D):³⁷ $F = M_0 f(D)$ and $f(D) = \exp(-D/d_1) + \exp(-D/d_2)$. Here, M_0 was a fitting parameter proportional to the magnetic moment of the standard microbead describing the bead heterogeneity. The two parameters d_1 and d_2 were two fitting parameters describing how rapid the force decays as

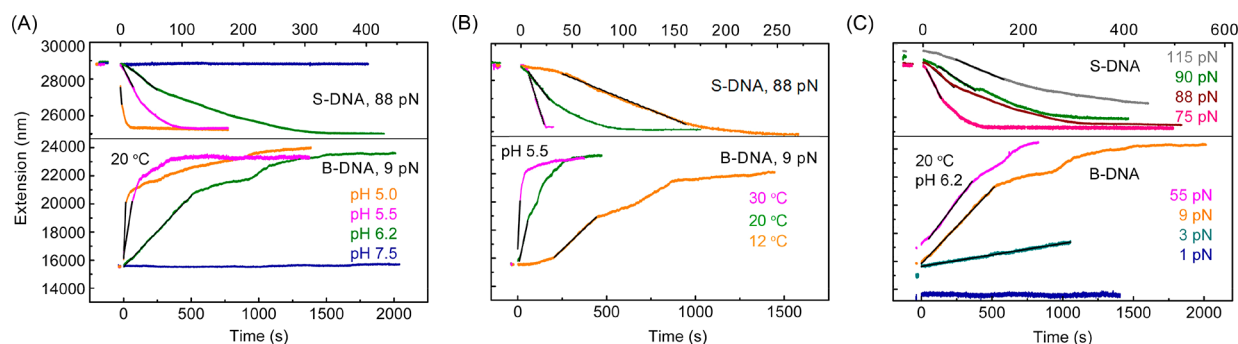


Figure 2. Dynamics of RecA polymerization on B-DNA (bottom panel) and S-DNA (upper panel) at various pH values (A), temperatures (B), and forces (C). The maximum polymerization rates on B-DNA and S-DNA obtained in the linear region after RecA was added (black lines) were calculated (Figure S3).

D increases, which were dependent only on the magnetic field and independent of the bead heterogeneity. Thus, $f(D)$ only was independent of the bead heterogeneity. For another microbead used in the experiment, the F as a function of D was calculated according to $F = M_i f(D)$, where M_i described the bead heterogeneity. The value of M_i was determined at the onset of overstretching transition for the end-closed and torsion-unconstrained DNA at 150 mM NaCl, pH 7.5, and 20 °C, where $F = 65$ pN.²⁰

A programmable syringe pump (New Era, NE-1000) was used to change buffer solutions inside the flow cell, and a home-built Peltier temperature controller was used to control the temperature inside the flow cell.⁶ We put the flow cell on the top side of the Peltier chip, and the bottom side of the Peltier chip was connected to 1 L of water through a heat-conducting plate. We changed the DC voltage to change the temperature differences between the two sides of the Peltier chip, and thus, the temperature of the flow cell was controlled. A thermocouple thermometer was used to monitor the temperature of the flow cell. After the DC voltage was changed, the temperature of the flow cell achieved the equilibrium in about 5 min.⁶

Proteins and Reaction Buffers. The *E. coli* RecA (UniProtKB P0A7G6) and human Rad51 (UniProtKB Q06609) were purchased from New England Laboratories and Abcam, respectively. The *Saccharomyces cerevisiae* Rad51 (UniProtKB P25454) was overexpressed in *E. coli* strain BL21 (DE3) competent cells that were transformed with pET30a-ScRad51. The cells were initially grown at 37 °C and then induced with 0.2 mM isopropyl- β -D-thiogalactoside (IPTG) for 16 h at 16 °C. Recombinant proteins were purified using an Ni Sepharose 6 Fast Flow purification kit according to the manufacturer's instructions (GE, USA). The full-length human replication protein A (RPA) composed of the subunits RPA1 (UniProtKB P27694), RPA2 (UniProtKB P15927), and RPA3 (UniProtKB P35244) was expressed and purified according to Henriksen et al.³⁸ The concentration of the purified protein samples was measured with a Pierce BCA protein assay kit (Thermo Scientific, USA).

Reaction buffers with 20 mM MES pH 5.0–6.2 or 20 mM Tris-HCl pH 7.5 containing 50 mM KCl, 10 mM MgCl₂, 1 mM DTT, and 1 mM ATP/ATP γ S were used in this study. In the RecA/Rad51 polymerization experiments, RecA and ScRad51 were diluted to 1 μ M and hRad51 was diluted to 300 nM with 1 mM ATP in the reaction buffer before they were added into the flow cell. In the DNA strand-exchange experiments, the RecA/Rad51-ssDNA filaments were prepared

by mixing the 12.5k nt single-ssDNA and RecA/Rad51 at a nucleotide/protein ratio of 3:1 at pH 7.5 with 1 mM ATP γ S and incubating at 37 °C for 15 min before being added into the flow cell.

Determination of the Nick-Free DNA Tethers. In our experiments, most DNA molecules contain nicks, and peeling may occur from those nicks during DNA overstretching. In order to generate a pure S-DNA without peeling, we need to find the tethered DNA molecules without nicks. As shown in Figure 1B, this task was achieved by measurements of the DNA force–extension curves through a stretching process containing a series of increasing constant forces (filled symbols), followed by a relaxing process containing the reverse series of decreasing constant forces (empty symbols) at the reaction buffer pH 5.5 and 20 °C. We chose those nick-free DNA molecules that underwent the nonhysteretic B-to-S transition indicated by the overlapping stretching and relaxing force–extension curves (olive) and discarded those nicked DNA molecules that partially underwent the hysteretic peeling transition indicated by the nonoverlapping stretching and relaxing force–extension curves (magenta). Even with a lower salt and higher temperature, 1 mM Tris-HCl, pH 7.5, and 30 °C, the nick-free DNA molecule underwent a hysteretic bubbling transition, generating a DNA bubble, which is shorter than S-DNA (orange).

In Figure 1B, the force–extension curve of S-DNA is fitted to a linear function of force (F): $27261 + 10.78F$ in the unit of nm. The predicted curves for ssDNA and DNA bubble (2ssDNA) are shown for comparison. The predicted curve for ssDNA was calculated using a monosalt-concentration-dependent empirical formula raised by Cocoo et al.,¹³ where a monosalt concentration of 1 M was used for the calculation. The predicted curve for 2ssDNA was calculated according to a picture that shows two ssDNA sharing the force evenly, where a monosalt concentration of 1 mM was used for the calculation using the same monosalt-concentration-dependent empirical formula.¹³

Studying Dynamics of RecA Polymerization and Formation of the RecA-Mediated DNA Triplex by DNA Extension Change. As shown in Figure 1C, S-DNA (~ 0.58 nm/bp) is longer than the RecA-dsDNA nucleoprotein filament as well as the RecA-mediated DNA triplex (~ 0.51 nm/bp), while B-DNA (~ 0.34 nm/bp) is shorter. Thus, we expect to observe an increase or a decrease in DNA extension, while RecA is polymerizing and the RecA-mediated DNA triplex is forming on B-DNA or S-DNA at constant forces, respectively. The rate of RecA polymerization and formation of

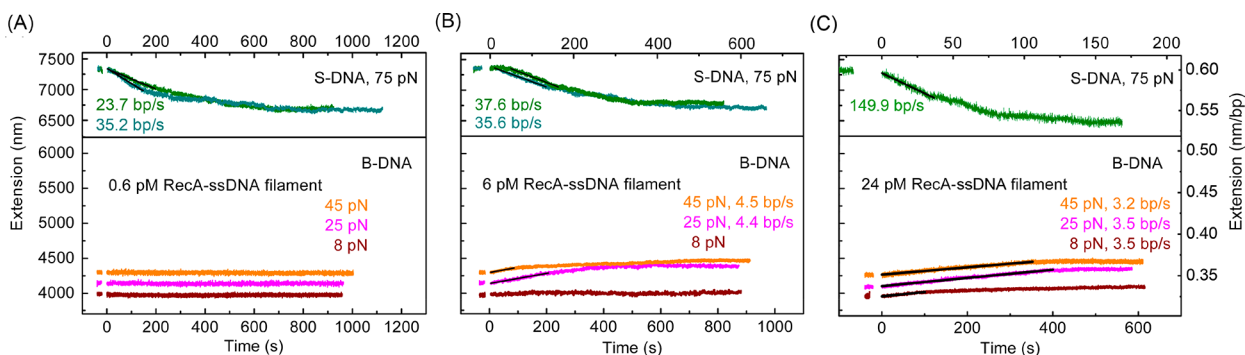


Figure 3. Formation of the RecA-mediated DNA triplex from B-DNA (lower panels) and S-DNA (upper panels) at pH 7.5 and 20 °C in the presence of the 0.6 (A), 6 (B), and 24 (C) pM RecA-ssDNA filament.

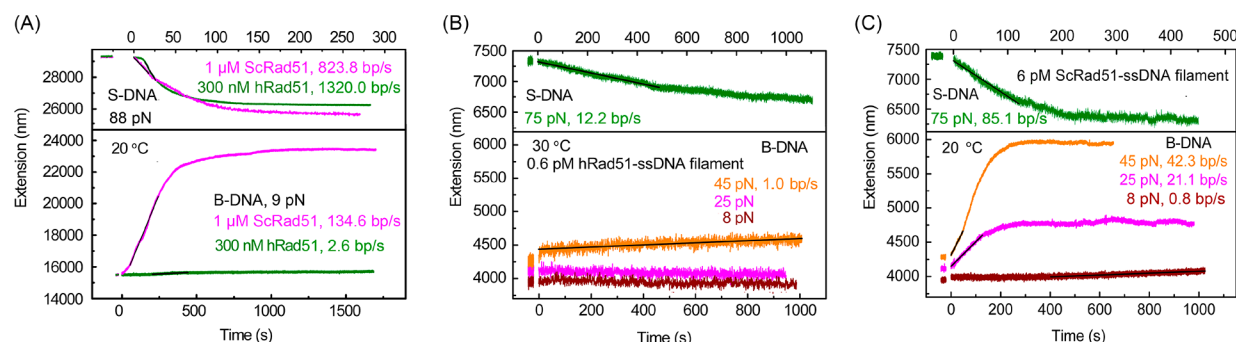


Figure 4. HRad51/ScRad51 polymerization and the formation of the hRad51/ScRad51-mediated DNA triplex. (A) Dynamics of hRad51/ScRad51 polymerization on B-DNA or S-DNA at pH 7.5. (B, C) Dynamics of the formation of the hRad51/ScRad51-mediated DNA triplex from B-DNA or S-DNA.

the RecA-mediated DNA triplex is calculated based on the extensions of these structures shown in Figure 1C. After RecA was added into the flow cell, we observed a region where the extension of DNA is changing fast and linearly (Figure 2). Beyond the linear region, the polymerization rate changed complicatedly. We selected the linear region and fitted the extension as a linear function of time, where the maximum polymerization rate was obtained.

RESULTS

RecA Polymerizes Faster on S-DNA than on B-DNA with the Same Buffer Conditions. As shown in Figure 2, we recorded and compared the dynamics of RecA polymerization on B-DNA and S-DNA at various pH values, temperatures, and constant forces. As shown in the lower panel of Figure 2A, we first studied RecA polymerization on B-DNA at pH 5.0–7.5 and 20 °C. It shows that the rate of RecA polymerization on B-DNA decreases with pH value, in agreement with previous studies.^{39,40} As shown in the upper panel of Figure 2A, the RecA polymerization experimental procedure was performed for S-DNA at a constant force of 88 pN at pH 5.0–7.5. The maximum rate of RecA polymerization on S-DNA in the linear region after RecA was added (black lines) was also calculated and plotted as a function of pH value (Figure S3A, filled squares), which also show a trend that RecA polymerizes faster at a lower pH on S-DNA. RecA polymerizes about 5-fold faster on S-DNA than on B-DNA at the same pH value.

As shown in Figure 2B, we next studied and compared the RecA polymerization on B-DNA and S-DNA at various temperatures of 12 °C (orange), 20 °C (olive), and 30 °C

(magenta) at pH 5.5. The maximum polymerization rates on B-DNA (empty squares) and S-DNA (filled squares) were plotted in Figure S3B as functions of temperature, which demonstrates that RecA polymerizes faster at higher temperatures and RecA polymerizes faster on S-DNA than on B-DNA at the same temperature.

Here we also studied the effects of force on RecA polymerization on B-DNA and S-DNA. As shown in Figure 2C, the time courses of DNA extension after RecA was added were recorded at various constant forces at pH 6.2 and 20 °C. The polymerization rate in the linear region after RecA was added was plotted in Figure S3C as a function of force, which shows that RecA polymerizes faster at higher forces for B-DNA (empty squares). However, RecA polymerizes slower at higher forces for S-DNA in the range 75–115 pN. Forces higher than 120 pN caused the depolymerization of the RecA-dsDNA filament (Figure S4).

We performed the RecA polymerization experiment using one molecule for each condition in Figure 2. In total, Figure 2 contains experimental data from 22 independent DNA molecules. In order to estimate the consistency of the rate of RecA polymerization at the same condition, we performed eight independent experiments at 55 and 80 pN, pH 6.2, and 20 °C (Figure S5). We found the rates of RecA polymerization on B-DNA and S-DNA were consistent with an error of less than 10% among independent molecules in our experiments.

Formation of the RecA-Mediated DNA Triplex from S-DNA. In order to study the strand exchange mediated by RecA, we stretched a nick-free DNA molecule at a constant force by magnetic tweezers and recorded the time courses of the extension after a freshly prepared RecA-ssDNA filament

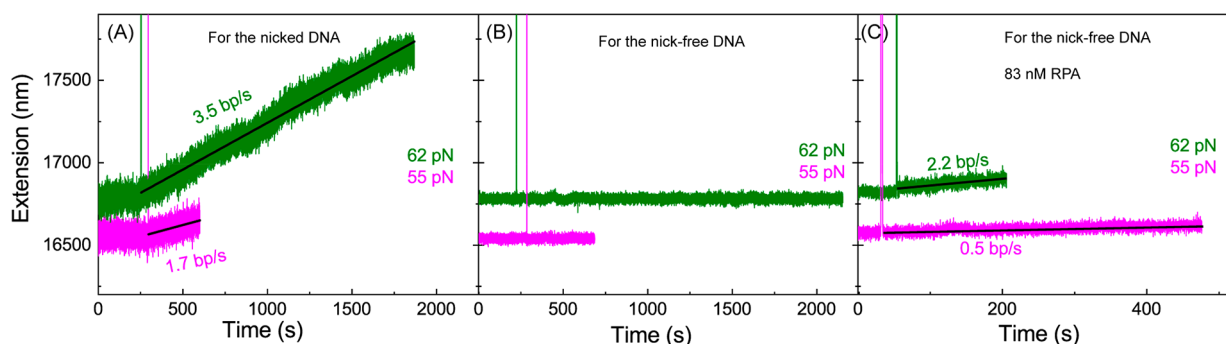


Figure 5. Polymerization of RecA on B-DNA stimulated by overstretching using a force pulse of 80 pN at pH 7.5 and 20 °C. The extension was fitted to linear functions of time (black lines), and the obtained rates were indicated. (A) The nicked DNA. (B) The nick-free DNA. (C) The nick-free DNA in the presence of RPA.

was added into the flow cell. If the ssDNA invades the dsDNA and forms a RecA-mediated DNA triplex (~ 0.51 nm/bp), we expect to observe a decrease in extension for S-DNA (~ 0.58 nm/bp) and an increase in extension for B-DNA (~ 0.34 nm/bp). As shown in Figure 3A, after a 0.6 pM 12.3k nt long RecA-ssDNA filament (concentration of 7.5 nM in RecA) was added to the flow cell, no significant increase in extension for B-DNA was observed (lower panel), while a decrease in the extension was observed for S-DNA (upper panel), which indicates the formation of the DNA triplex from S-DNA but not B-DNA at this low RecA-ssDNA filament concentration. At higher RecA-ssDNA filament concentrations of 6 and 24 pM, the DNA triplex formed from both S-DNA and B-DNA (Figure 3B,C). The rate of DNA triplex formation was calculated in the linear region after the filament was added (black lines), which shows that the RecA-mediated DNA triplex forms faster from S-DNA than from B-DNA. Note that RecA did not polymerize on B-DNA or S-DNA at pH 7.5 and 20 °C (blue, Figure 2A and Figure S6), and thus, the extension change in Figure 3 cannot be caused by the free RecA polymerization on DNA.

Rad51 Polymerization on S-DNA and Formation of the Rad51-Mediated DNA Triplex on S-DNA. Here we also studied the hRad51/ScRad51 polymerization and the formation of the hRad51/ScRad51-mediated DNA triplex using similar procedures as the above RecA experiments. As shown in Figure 4A, the dynamics of hRad51 (olive) and ScRad51 (magenta) polymerization on S-DNA (upper panel) and B-DNA (lower panel) were recorded and compared at pH 7.5 and 20 °C. The polymerization rate in the linear region after Rad51 was added was calculated and indicated in the figure. The hRad51 polymerization rate on S-DNA is 1320 bp/s at 88 pN, much greater than 2.6 bp/s on B-DNA at 9 pN. The ScRad51 polymerization rate on S-DNA is 823 bp/s at 88 pN, which is also greater than 134 bp/s on B-DNA at 9 pN.

As shown in Figure 4B,C, we next studied the formation of the hRad51/ScRad51-mediated DNA triplex from B-DNA and S-DNA. After a 0.6 pM hRad51-ssDNA filament (7.5 nM in hRad51 concentration) or 6 pM ScRad51-ssDNA filament (75 nM in ScRad51 concentration) was added in the flow cell, while the DNA molecule was stretched with a constant force, we observed a region where the DNA extension was changing linearly (black lines). The rate of the formation of the Rad51-mediated DNA triplex in the linear region was calculated and indicated in the figures, which demonstrated that hRad51/ScRad51-mediated DNA forms faster from S-DNA than from B-DNA with the same buffer conditions. Note that we did not

observe hRad51/ScRad51 polymerization at the low concentration of 7.5 nM free hRad51 or 75 nM free ScRad51 (Figure S7), and thus, the change in DNA extension in Figure 4B,C was not caused by hRad51 and ScRad51 polymerization.

DISCUSSION

In this study, we have studied the dynamics of RecA polymerization on B-DNA and S-DNA at various pH values, temperatures, and constant forces with single-molecule magnetic tweezers (Figure 2). We found RecA polymerizes faster on S-DNA than B-DNA with the same buffer conditions. A low pH and high temperature accelerate RecA polymerization on both B-DNA and S-DNA, which is in agreement of previous results of RecA polymerization on B-DNA.^{39–46} One difference between our results and previous results is that at pH 7.5 we did not see RecA polymerization on S-DNA, while previous studies have shown RecA could polymerize on overstretched DNA²⁷ and RecA could polymerize on B-DNA with the assistant of DNA overstretching.³⁰ We assume that in previous studies end-opened DNA tethers were used and RecA nucleated on the peeled ssDNA generated at nicks or open ends during DNA overstretching. This assumption was verified that at pH 7.5 RecA could polymerize on B-DNA with the assistant of DNA overstretching for nicked DNA but not for nick-free DNA (Figure 5).

We found an increase in force led to an increase in the RecA assembly rate on B-DNA, but decreased the RecA assembly rate on S-DNA (Figure 2C, Figure S3C). Forces even larger than 120 pN can accelerate the depolymerization of RecA from S-DNA (Figure S4). This phenomenon can be explained by the opposite energy contributions of the applied force during RecA polymerization on B-DNA and S-DNA. The extension of DNA increases during RecA polymerizing on B-DNA, and thus the applied force makes a positive contribution to RecA polymerization. However, the extension of DNA decreases during RecA polymerizing on S-DNA, and thus the applied force makes a negative contribution to RecA polymerization. Considering the molecule structure of S-DNA is still unclear, we do not have a clear molecule explanation of why RecA polymerizes slower at larger forces on S-DNA. In our RecA polymerization experiments, the 48.5k bp long λ DNA was used, and the dynamics of extension we observed was likely caused by elongation of RecA-dsDNA filaments started from multiple nucleated sites.

Previous magnetic tweezer experiments have revealed that negative supercoiling helped the formation of the RecA-mediated DNA triplex.^{32,33} In our experiments, torsion-

unconstrained DNA constructs were used and thus our DNA molecules were not negatively supercoiled. We found the formation of the RecA-mediated DNA triplex from the less helical S-DNA (about 10° twist per bp)²³ was faster than from B-DNA (about 35° twist per bp) with homologous RecA-ssDNA filaments (Figure 3). Particularly in the presence of the low concentration of 0.6 pM RecA-ssDNA, the formation of the RecA-mediated DNA triplex was only observed from S-DNA but not from B-DNA.

For the Rad51/RecA family eukaryotic recombinases ScRad51 and hRad51, similar polymerization and strand-exchange results for S-DNA were obtained (Figure 4). hRad51 and ScRad51 polymerase faster on S-DNA than B-DNA with the same buffer conditions (Figure 4A). The formation of both hRad51-mediated and ScRad51-mediated DNA triplex was also faster from S-DNA than from B-DNA (Figure 4B,C).

For end-closed and torsion-unconstrained DNA, it was reported that small DNA bubbles may form at AT-rich regions coexisting with S-DNA formed at GC-rich regions in the presence of replication protein A (RPA).²¹ Thus, one may be concerned that the faster assembly rate and triplex formation rate on the overstretched DNA could result from the ssDNA exposure to these small bubbles. In order to address this concern, we studied the polymerization of RecA on B-DNA stimulated by overstretching using a force pulse of 80 pN at pH 7.5 and 20 °C (Figure 5). For the nicked DNA, RecA could polymerize on B-DNA with the assistant of DNA overstretching, which can be attributed to the nucleation of RecA on the peeled ssDNA formed at the DNA nicks during DNA overstretching (Figure 5A). For nick-free DNA, however, RecA could not polymerize on B-DNA even with the assistant of DNA overstretching (Figure 5B). Thus, it was unlikely that ssDNA of the DNA bubble formed during overstretching of the nick-free, end-closed, and torsion-constrained DNA in this buffer condition. For the nick-free DNA and in the presence of 83 nM RPA, which specifically binds to ssDNA, RecA could polymerize on B-DNA with the assistant of DNA overstretching (Figure 5C), which can be explained by RecA nucleated on the exposed ssDNA of the DNA bubbles formed during DNA overstretching in the presence of RPA.

Furthermore, we have made an 8k bp, GC-rich, end-closed, and torsion-unconstrained DNA where DNA bubbles were unlikely to form during overstretching due to its high GC sequences (Figure S1D,E). Using the GC-rich DNA, we find the Rad51 assembles faster and the DNA triplex forms faster on S-DNA than B-DNA as well (Figure 6). Thus, the faster assembly rate and triplex formation rate on the overstretched DNA are unlikely to be resulted from the ssDNA exposed to these small bubbles.

Without binding partners, S-DNA only exists under a high tension of greater than 60 pN, and thus S-DNA is unlikely to exist at a larger scale *in vivo*. However, if S-DNA exists with the assistant of DNA intercalators and other DNA-distorting proteins in cells, S-DNA is vulnerable to be captured by Rad51 and RecA proteins or Rad51/RecA-ssDNA filaments due to the faster kinetics compared to their interactions with B-DNA. In all, our results provide evidence that S-DNA can directly interact with RecA and Rad51 *in vitro* and shed light on the possible functions of S-DNA.

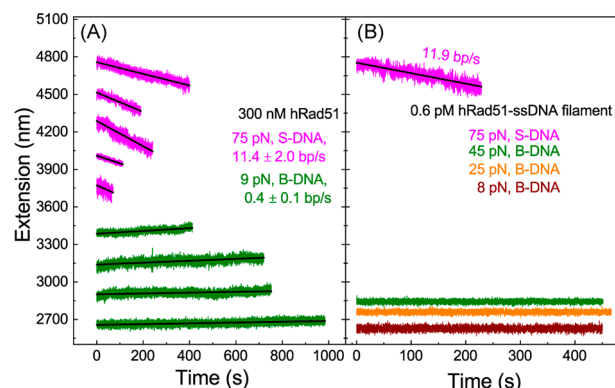


Figure 6. HRad51 polymerization and the formation of the hRad51-mediated DNA triplex for the 8k bp GC-rich, end-closed, and torsion-unconstrained DNA at pH 7.5 and 20 °C. (A) The polymerization of hRad51 on B-DNA at 9 pN (olive) and on S-DNA at 75 pN (magenta). The DNA extension during polymerization at the linear regions was fitted to linear functions of time (black lines), and the average and standard deviation in the polymerization rate was calculated. The time courses were shifted for clarity. (B) The formation of the hRad51-mediated DNA triplex on B-DNA and on S-DNA in the presence of a 0.6 pM hRad51-ssDNA filament. The rate of formation of the DNA triplex on S-DNA was obtained by fitting the time course to a linear function of time (black line). We did not observe a significant increase in extension for the B-DNA.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.biochem.8b01125.

Preparation of DNA constructs, extension vs time plots, maximum rate of RecA polymerization, and time courses (PDF)

Accession Codes

Escherichia coli RecA: P0A7G6. Human Rad51: Q06609. *Saccharomyces cerevisiae* Rad51: P25454. Human RPA1: P27694. Human RPA2: P15927. Human RPA3: P35244.

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Author Contributions

X.H.Z. designed this work. X.C.Z., H.F., L.S., Y.Y.J., E.C.Z., G.X.L., X.F.C., Z.L., and W.Q.W. performed the experiments. X.H.Z. and X.C.Z. wrote the manuscript.

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Notes

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ABBREVIATIONS

E. coli, *Escherichia coli*; hRad51, human Rad51; ScRad51, *Saccharomyces cerevisiae* Rad51; ssDNA, single-stranded DNA; dsDNA, double-stranded DNA; bp, base pair; nt, nucleotide

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